

# index\_engine — Methodology & Banding Study

A configurable equity-index construction & reconstitution engine  
Generated 2026-06-24 05:20. Methodology is general; the FTSE Russell U.S. family is the reference configuration. Empirical figures below come from a synthetic universe run through the real ranking engine.



## 1. The construction process

An equity index is rebuilt on a fixed schedule by passing the investable universe through an ordered set of rules. The engine implements each as a discrete, testable stage:

**Eligibility screening.** Remove what should not be in a common-equity index: sub-\$1 prices, sub-threshold market caps, non-U.S. domicile, and non-operating structures (ETFs, funds, preferred, warrants/rights/units, SPACs, LPs and royalty trusts). The reference config uses a \$1.00 price floor and a \$30M cap floor.

**Capitalization ranking.** Rank survivors by total market capitalization, descending. Multi-class issuers (e.g. a Class A / Class C pair) are consolidated so the company's full size determines its rank, while only the primary class is included.

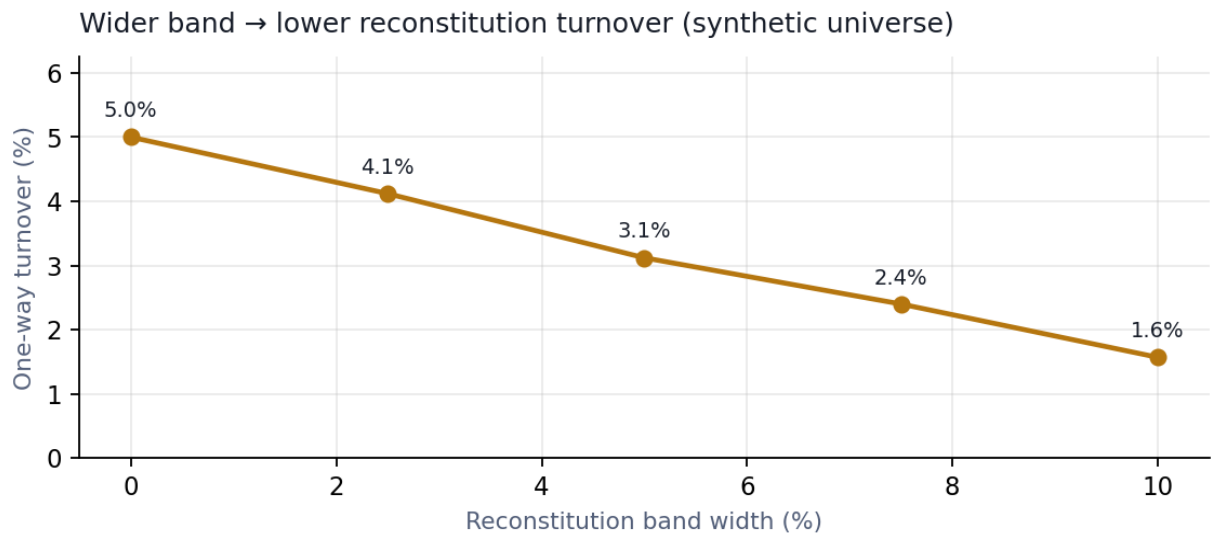
**Banding / buffer.** Rather than a hard rank cutoff, an existing member is retained while its rank stays inside an upper buffer, and a new entrant must clear a tighter lower threshold. This hysteresis is the turnover-control lever studied in section 2.

**Tier split.** The broad index is partitioned into a large-cap and a small-cap tier at a rank/capitalization breakpoint, itself banded so names near the line do not migrate every period.

**Float weighting.** Members are weighted by free-float-adjusted market cap, so each tier's weights sum to 100% and the weight reflects investable shares, not total shares outstanding.

## 2. Why banding exists: a turnover study

Banding trades index-size precision for stability. To quantify it, a synthetic universe is ranked to form a prior index, then re-ranked after one year of simulated price drift. Only the band width varies between runs — every other step is the production engine. One-way turnover is the average of additions and deletions as a share of index size.



| Band width | Additions | Deletions | Index size | One-way turnover |
|------------|-----------|-----------|------------|------------------|
| 0.0%       | 150       | 150       | 3000       | 5.00%            |
| 2.5%       | 121       | 126       | 2995       | 4.12%            |
| 5.0%       | 91        | 96        | 2995       | 3.12%            |
| 7.5%       | 68        | 76        | 2992       | 2.40%            |
| 10.0%      | 48        | 46        | 3002       | 1.57%            |

**Finding.** Widening the band from 0% to 10% lowers one-way turnover from 5.00% to 1.57% — a 69% relative reduction — while holding index size within a few names of target. Lower turnover means fewer forced trades at reconstitution and lower implied transaction costs, which is precisely the tradeoff banding is designed to manage. Figures are illustrative (synthetic universe, simulated returns); the point is the monotonic relationship and the mechanism, not the exact percentages.

### 3. Engine & validation

The methodology stages are pure, deterministic transforms over a securities table, covered by a network-free test suite:

| Area          | What the tests assert                                                                     |
|---------------|-------------------------------------------------------------------------------------------|
| Eligibility   | each filter (price, cap, exchange, suffix, SPAC, ETF, currency) excludes correctly        |
| Consolidation | multi-class caps are summed; only the primary class is the inclusion unit                 |
| Ranking       | rank is dense, 1-based, and ordered by descending capitalization                          |
| Banding       | existing members inside the buffer are retained; new entrants beyond the cutoff are gated |
| Tiers         | large/small tiers are disjoint and nest within the broad index                            |
| Weighting     | float-adjusted weights sum to 100%                                                        |

Live data sourcing (NASDAQ Trader universe, market data, SEC EDGAR XBRL) is implemented in the fetch modules and runs against real endpoints with caching and rate-limit handling; this report stays offline and synthetic so the methodology and its behaviour can be reproduced anywhere.